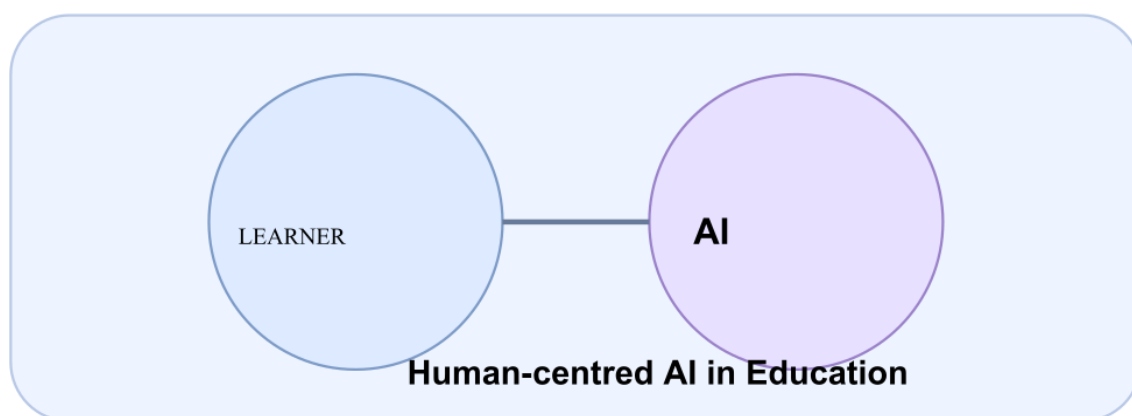


ARTIFICIAL INTELLIGENCE IN EDUCATION

Opportunities, responsible use, competencies and a practical classroom workflow



Executive overview

Artificial intelligence is changing how learners search for information, practise skills, create content and receive feedback. The strongest educational approach is not to replace human judgement with AI, but to use AI as a support layer while keeping teachers, learners and institutions responsible for decisions.

UNESCO's guidance emphasizes a human-centred approach that protects human agency, inclusion, equity, privacy and cultural diversity. Its student competency framework organizes AI capabilities around a human-centred mindset, ethics, AI techniques and applications, and AI system design. ■cite■turn0search0■turn0search1■

Report at a glance

Area	Potential value	Main caution
Learning	Personalized explanations, practice and feedback	Over-reliance or incorrect output
Teaching	Lesson planning, examples and differentiation	Quality control and workload shifts
Assessment	Feedback and formative support	Academic integrity and validity
Creativity	Drafting, ideation and multimodal creation	Attribution, copyright and bias

1. HOW AI CAN SUPPORT LEARNING

AI can provide fast, adaptive assistance across different stages of learning. Used well, it can help students explore concepts, compare explanations, practise with examples and receive feedback. However, outputs should be treated as material to evaluate rather than unquestioned facts. UNESCO notes that GenAI can create convincing outputs without genuinely understanding the real world, making critical evaluation especially important.

■cite■turn0search15■



A practical human-in-the-loop workflow

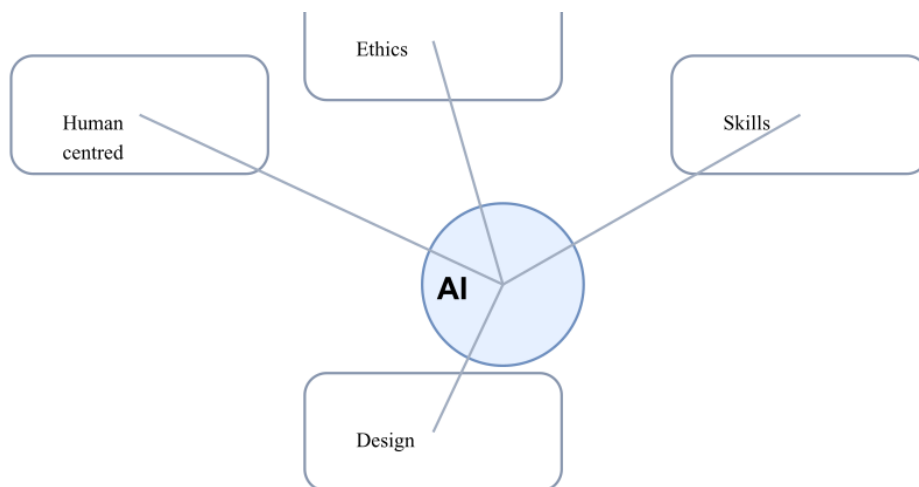
1. Define the learning goal. 2. Ask AI for a first explanation or set of examples. 3. Check the response against reliable sources, course material or teacher guidance. 4. Improve, challenge or correct the output. 5. Produce the final work using the learner's own reasoning.

Example classroom uses

Use case	AI can assist with	Human role
Concept tutoring	Explain a difficult topic at multiple levels	Check accuracy and choose the best explanation
Practice	Generate questions, hints and worked examples	Attempt independently and verify solutions
Writing	Brainstorm, outline and suggest revisions	Own the argument, evidence and final wording
Coding	Explain errors and propose alternatives	Test code, understand it and make final decisions

2. AI COMPETENCIES & RESPONSIBLE USE

AI literacy is broader than knowing how to write prompts. Learners need to understand what AI systems can and cannot do, recognize ethical risks, evaluate outputs, and make responsible decisions. UNESCO's 2024 student framework identifies 12 competencies across four dimensions and three progression levels: Understand, Apply and Create. [cite turn0search1](#)



Core competency areas

Dimension	What it means in practice
Human-centred mindset	Keep human needs, agency and judgement at the centre.
Ethics of AI	Recognize bias, privacy, safety, fairness and accountability issues.
AI techniques & applications	Understand basic AI concepts and how different systems are used.
AI system design	Think about how AI systems are created, tested and improved.

Responsible-use checklist

- Verify important factual claims before relying on them.
- Do not enter sensitive personal information into AI tools unless appropriate safeguards are in place.
- Be transparent about meaningful AI assistance when required by an institution or assignment.
- Treat generated content as a draft or aid when the learning objective is independent reasoning.
- Check for bias, missing perspectives and culturally inappropriate assumptions.

3. BENEFITS, RISKS AND A PRACTICAL POLICY

The educational value of AI depends on context. UNESCO recommends policy approaches that protect human agency, promote inclusion and equity, monitor and validate AI systems, build AI competencies, and support teachers and researchers. ■cite■turn0search2■turn0search3■

Decision table

Question	If YES	If NO
Does AI support the learning objective?	Use it with clear boundaries.	Avoid it or redesign the task.
Can the output be verified?	Check it using trusted material.	Treat it as uncertain and seek another source.
Is personal data involved?	Use approved privacy safeguards.	Proceed with normal caution.
Does the task require original reasoning?	Use AI for support, not substitution.	AI may have a larger role if permitted.
Is disclosure required?	State how AI was used.	Follow the institution's normal rules.

Key takeaway

AI works best in education when it expands what people can do without removing the responsibility to think. A strong model is **AI assistance + human verification + learner ownership**. UNESCO's current education work similarly emphasizes human agency, critical thinking, ethics, inclusion and equitable technology use. ■cite■turn0search4■turn0search5■

Sources

1. UNESCO, *Guidance for generative AI in education and research*. 2. UNESCO, *AI competency framework for students*. 3. UNESCO, *AI competency framework for teachers*. 4. UNESCO Digital Education resources.